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Fifteenth International Linguistics Olympiad

Dublin (Ireland), 31 July – 4 August 2017

Individual Contest Problems

Rules for writing out the solutions

Do not copy over the problems. Write down your solution to each problem on a separate sheet or sheets. On each sheet indicate the number of the problem, the number of your seat and your surname. Otherwise your work may be mislaid or misattributed.

Your answers must be well-supported by argument. Even a perfectly correct answer will be given a low score unless accompanied by an explanation.

Problem 1 (20 points). Here are some arithmetic equalities in Birom:

1. **tùŋūn**^{2} + **tàt** + **nààs** = **bākūrū bībā ná vɛ̀ rwīīt**
2. **tàt** $^{\textbf{nààs}}$ = **bākūrū bītīīmìn ná vɛ̀ ʃāātàt**
3. **tàāmà**^{2} + **ʃāātàt** + **gwīnìŋ** = **bākūrū bīnāās ná vɛ̀ ʃāāgwīnìŋ**
4. **ʃāātàt** $^{\textbf{gwīnìŋ}}$ = **ʃāātàt**
5. **rwīīt**^{2} + **bà** + **tùŋūn** = **bākūrū bītūŋūn ná vɛ̀ ʃāāgwīnìŋ**
6. **bà** $^{\textbf{tùŋūn}}$ = **bākūrū bībā ná vɛ̀ rwīīt**
7. **ʃāātàt**^{2} + **nààs** + **tàt** = **bākūrū bītāāmà ná vɛ̀ nààs**
8. **nààs** $^{\textbf{tàt}}$ = **bākūrū bītūŋūn ná vɛ̀ nààs**
9. **kūrū ná vɛ̀ nààs** + **kūrū ná vɛ̀ ʃāātàt** = **kūrū ná vɛ̀ tìīmìn** + **bà** + **kūrū ná vɛ̀ tùŋūn**

All numbers in this problem are greater than 0 and less than 125.

(a) Write the equalities (1–9) in numerals.

(b) Write the numbers **bākūrū bītāt**, **ʃāāgwīnìŋ**, **kūrū** and the equalities (A) and (B) in numerals.

A. **bākūrū ʃāābītāt** − **tàt** − **kūrū ná gwɛ̄ gwīnìŋ** = **bākūrū bītāāmà ná vɛ̀ rwīīt**

B. **bākūrū bīnāās ná gwɛ̄ gwīnìŋ** − **kūrū ná vɛ̀ bà** − **kūrū ná vɛ̀ tàt** = **kūrū ná vɛ̀ rwīīt**

(c) Write out in Birom: 6, 22, 97, 120.

⚠ Birom belongs to the Plateau group of the Benue-Congo branch of the Atlantic-Congo family. It is spoken by approx. 1,000,000 people in Nigeria.

ɛ is a vowel. **ŋ** and **ʃ** are consonants. The doubling of a vowel denotes length. The marks ˊ, ¯ and ˋ indicate high, middle, and low tone, respectively.

$a^k = \underbrace{a \times a \times \cdots \times a}_{k \text{ times}}; a^1 = a.$ —*Milena Veneva*